

Undecidable Problem: The word problem for string rewriting systems

Statement of the problem: A string rewriting system, also called a Thue system, consists of a finite alphabet Σ & a finite set of rules of the form $u \leftrightarrow v$, where u & v are strings over Σ .

The Word Problem: Given a Thue system & two strings X and Y over Σ , can X be transformed into Y by a finite sequence of rule applications?

Why this problem is not decidable: All possible sequences of rule applications can be generated ~~sysm~~ systematically. If X can be transformed into Y , this search eventually finds a valid sequence. If it cannot, the search may never end and no point is ever reached at which failure can be concluded. The problem is therefore recursively enumerable but not recursive.

History:

@ Axel Thue: The Norwegian mathematician introduced these rewriting systems and asked whether a general procedure exists to decide whether one string can be transformed into another. Since no formal definition of an algorithm existed at the time, the question could not be answered rigorously.

⑥ Formalisation of Computability: The work of Alonzo Church, Alan Turing gave a precise definition of an effective procedure. This made it possible to prove that certain problems admit no algorithm.

⑦ Emil Post & Andrey Markov: Both proofs use a reduction from the halting problem → ① A configuration of a Turing machine, consisting of its tape contents, current state and head position, is encoded as a string.

② Each transition of the machine is encoded as a rewriting rule.

③ Under the encoding, the machine halts if & only if the string for its initial configuration can be rewritten into a string representing a halting configuration.

An algorithm for the Word Problem would therefore yield an algorithm for the halting problem, which is known not to exist. Hence the Word Problem is undecidable.